Functional Requirements Document – Notification

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (Sub-module: Communication → Notification, incl. deadline reminder engine, code `NOTIF`) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-Notification.md v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related Documents

Document	Purpose	Location
BRD – Notification	Business needs this FRD details (`REQ-NOTIF-nn`, `BR-NOTIF-nn`, `AC-NOTIF-nn`)	`../../../BRD/04-Communication/04.2-Notification/BRD-Notification.md`
FRD – Overall	Roles, global rules, Authentication, Legal, module index	`../../00-Overall/FRD-TaskFlow-Overall.md`
FRD – Mail (sibling)	Receives reminder mails in the Inbox	`../04.1-Mail/FRD-Mail.md`
FRD – Reminder Settings	Reminders on/off, deadline-day reminder, "Also deliver reminders to Mail", default reminder days	`../../06-Settings/06.1-Reminder-Settings/FRD-Reminder-Settings.md`
FRD – Application Settings	In-app notifications toggle, application timezone	`../../06-Settings/06.2-Application-Settings/FRD-Application-Settings.md`
FRD – Task Management	Task reminder fields, recipients, task events, task details screen	`../../02-TaskFlow-Monitoring/02.1-Task-Management/FRD-Task-Management.md`
Development Plan	Screen 12 Notification Center	`TF - Documents/Screens Wise Feature Development Plan/12-notification-center-screen.md`
HLD / LLD	Technical design of Notification	`../../../HLD/04-Communication/04.2-Notification/HLD-Notification.md`, `../../../LLD/04-Communication/04.2-Notification/LLD-Notification.md`

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Notification sub-module: the Notification Center screen and header bell, the event notifications written when tasks and accounts change, and the automatic deadline reminder engine that sends first, second, due-today and overdue reminders exactly once.

1.2 Scope

In scope: everything listed in BRD-Notification §3.1. Out of scope (BRD-Notification §3.2): editing reminder switches and defaults (Reminder Settings FRD), setting a task's reminder days and recipients (Task Management FRD), outside email or SMS, deleting or archiving notifications, live push, status-change notifications, catching up past days and tracking failed deliveries.

Change from original BRD

Original BRD item	Current requirement
Screen 12 "Access: All registered users"	Access by `notifications.view`, granted by default to Administrator, Manager, Employee (FR-NOTIF-01).
Screen 12 "Unread notification count"	The bell shows a dot while anything is unread; no number is printed (FR-NOTIF-03).
Screen 12 "Notification filters"	Two filters: Type and Read state (FR-NOTIF-09, FR-NOTIF-10).
Screen 12 FR 5 "Create due-today alerts"	Due-today reminders use the type "Deadline reminder" and the text "… is due today at *HH:mm*." (FR-NOTIF-39, FR-NOTIF-45, FR-NOTIF-46).
Section 3.2 fixed schedule (two days / one day before)	Per-task first and second reminder days (0–30, defaults 2 and 1) and a Daily or Weekly overdue frequency (FR-NOTIF-36 – FR-NOTIF-40).
Section 3.2 "Deadline changed → recalculate future reminders"	Every run uses the task's current deadline and status (FR-NOTIF-44, FR-NOTIF-50).
Section 3.4 "Email Notifications" via SMTP	Reminders are also placed in the internal Mail Inbox when "Also deliver reminders to Mail" is on (FR-NOTIF-47).
Section 3.5 history "processing status, delivery attempt details"	History records each delivered reminder with status "Sent"; attempts and failures are not tracked (FR-NOTIF-48).
Not in original BRD	Notifications for reassignment, comments, worked hours and account security events (FR-NOTIF-23 – FR-NOTIF-28).

1.3 Audience

Developer, tester, reviewer and product owner of TaskFlow.

1.4 Conventions

- Requirement IDs: `FR-NOTIF-nn`. Use cases: `UC-NOTIF-nn`. BRD items: `REQ-NOTIF-nn`, sub-module rules `BR-NOTIF-nn`, global rules `BR-nn` (overall BRD), NFRs `NFR-NOTIF-nn`, acceptance criteria `AC-NOTIF-nn`.
- Priority: **Must / Should / Could** (MoSCoW), taken from the traced BRD requirement.
- "The system shall …" = mandatory behaviour. Text in *italics* inside messages is a value filled in at run time.
- "Today", "day" and dates in reminder texts are in the application timezone.

1.5 Definitions & Abbreviations

Term	Meaning
Notification	A short in-app alert for one user; scope is always **Own**

Term	Meaning
Event notification	A notification caused by a change: assignment, reassignment, comment, worked hours, account event
Reminder engine	Background process that checks open tasks at start-up and every 60 seconds and sends deadline reminders
First / second reminder	Reminder sent exactly the task's first / second reminder days (0–30) before the deadline day
Due-today reminder	Reminder sent on the deadline day before the deadline time
Overdue reminder	Reminder sent after the deadline day, Daily or Weekly, while the task is open
Reminder recipients	The task's notification recipients (the assignee is always one) plus the task creator, active users only
Reminder history	Record of delivered reminders, one per task, person, reminder type and day
Open task	A task that is not completed, not cancelled and not deleted
Application timezone	The single timezone set in Application Settings (UTC when not set)

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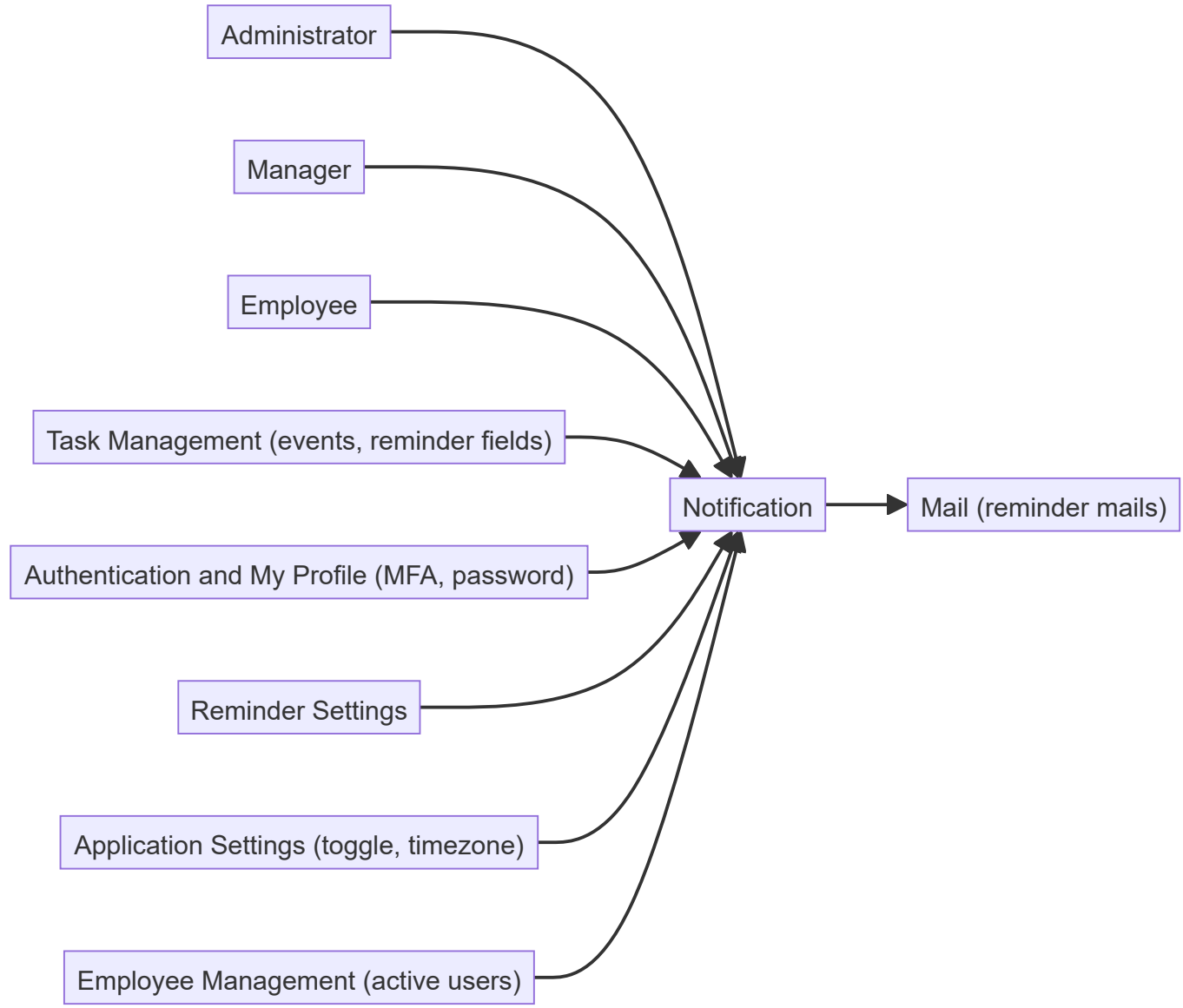
2. System Overview

2.1 Module Map

Code	Module	Child modules	Primary roles
NOTIF	Communication → Notification	Notification Center (with header bell), Event notifications, Deadline reminder engine	Administrator, Manager, Employee, Custom Role (as granted); System

Other modules are described in the overall FRD (`../../00-Overall/FRD-TaskFlow-Overall.md`).

2.2 Context Diagram



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3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	System administrator; grants `notifications.*` keys to custom roles; receives own notifications	Own notifications only
Manager	Leads a team or project; receives reminders for tasks created	Own notifications only
Employee	Works on assigned tasks; receives alerts as assignee or recipient	Own notifications only
Custom Role	Role created by an Administrator	Own notifications, as granted
System	Reminder engine and event sources; no user permission	Writes notifications for affected users

No role, including Administrator, sees another user's notifications (BR-NOTIF-01).

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Notification › Open the Notification Center, header bell and unread dot	`notifications.view`	Granted	Granted	Granted	Configurable
Notification › Mark one or all notifications as read	`notifications.manage`	Granted	Granted	Granted	Configurable
Reminder Settings › Reminder switches (controls the engine; specified in the Reminder Settings FRD)	`settings.reminders.manage`	see Reminder Settings FRD			
Application Settings › In-app notifications toggle, timezone (specified in the Application Settings FRD)	`settings.application.manage`	see Application Settings FRD			

A new custom role starts with neither `notifications.*` key. The reminder engine and event notifications need no user permission.

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4. Functional Requirements by Module

4.1 Module: Notification (NOTIF)

Purpose: Tell each user about their own work in one place and send deadline reminders automatically, once each. · **BRD reference:** BRD-Notification §5 SCREEN 1, workflows 4.1 – 4.3; BR-NOTIF-01 – BR-NOTIF-13; global rules on notifications, reminders and timezone.

4.1.1 Screen: Notification Center (with header bell)

Field	Value
Route / location	`/notifications` (menu: Communication Management → Notifications); header bell on every page
Access	`notifications.view` (Administrator, Manager, Employee by default; Custom Role if granted); marking read needs `notifications.manage`
Entry points	Side menu item "Notifications"; bell icon in the header

Screen elements

Element	Type	Description	Mandatory	Visible to
Header bell	Icon link	Opens the Notification Center; dot while anything is unread	–	`notifications.view`
Title "Notifications"	Heading	Page title	–	`notifications.view`
Range label	Text	"1–20 of 45 notifications"; adds "match the current filters" when filtered	–	`notifications.view`
Mark all as read	Button	Marks every unread notification read; disabled when nothing is unread	–	`notifications.manage`
Type	Drop-down	All types, Task assigned, Task reassigned, Deadline reminder, Overdue, System	No	`notifications.view`
Read state	Drop-down	All, Unread, Read	No	`notifications.view`

Element	Type	Description	Mandatory	Visible to
Reset	Button	Clears both filters	–	`notifications.view`
Notification row	List row	Type chip, date and time, message, task code link; unread rows highlighted	–	`notifications.view`
Mark as read	Button (unread rows)	Marks the row read	–	Shown to every viewer; use needs `notifications.manage`
Previous / Next, "Page X of Y"	Paging	20 per page	–	`notifications.view`
Empty / no-match / error states	Panels	"No notifications yet"; "No notifications match these filters" with Clear filters; error with Retry	–	`notifications.view`

Functional requirements

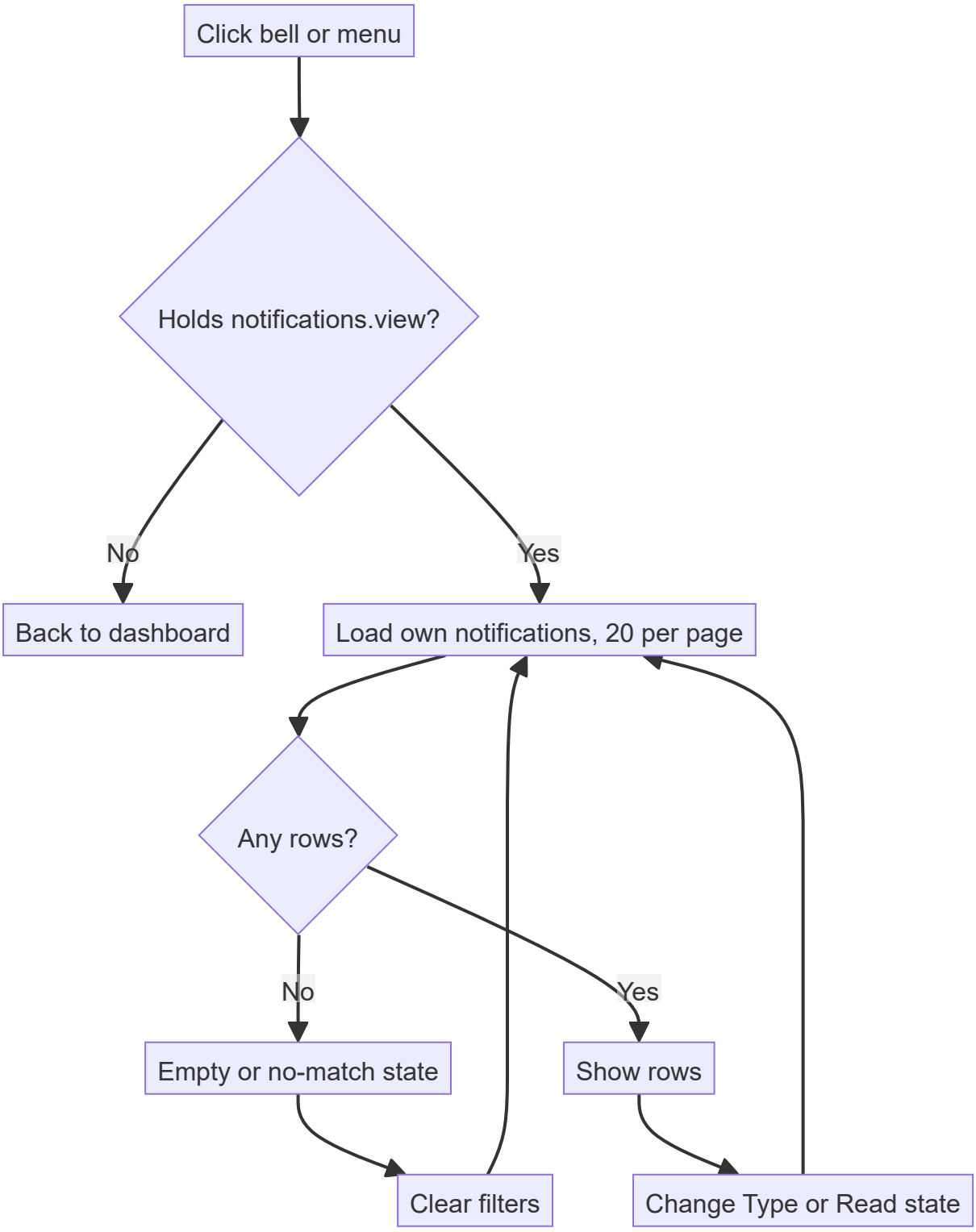
ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-01	The system shall open `/notifications` only for a user holding `notifications.view` and shall return any other user to their dashboard without a message.	Must	`notifications.view`	REQ-NOTIF-01
FR-NOTIF-02	The system shall show a bell in the header of every page to users holding `notifications.view`, linking to the Notification Center.	Must	`notifications.view`	REQ-NOTIF-02
FR-NOTIF-03	The system shall show a dot on the bell while the user has at least one unread notification and none otherwise, refreshing the unread state on every page change, every 15 seconds and when the browser tab becomes visible again.	Must	`notifications.view`	REQ-NOTIF-03, NFR-NOTIF-02
FR-NOTIF-04	The system shall list only the signed-in user's own notifications, newest first.	Must	`notifications.view`	REQ-NOTIF-04, BR-NOTIF-01
FR-NOTIF-05	The system shall show 20 notifications per page with Previous, Next, "Page X of Y" and a range label such as "1–20 of 45 notifications", adding "match the current filters" when a filter is active.	Should	`notifications.view`	REQ-NOTIF-05
FR-NOTIF-06	The system shall accept a requested page size from 1 to 100 and use 20 for any value outside that range, without a message.	Should	`notifications.view`	REQ-NOTIF-05, NFR-NOTIF-03
FR-NOTIF-07	The system shall show on each row a type chip, the date and time, the message and, when the related task still exists and is not deleted, the task code as a link.	Must	`notifications.view`	REQ-NOTIF-06

ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-08	The system shall highlight unread rows.	Should	`notifications.view`	REQ-NOTIF-07
FR-NOTIF-09	The system shall filter the list by Type: All types, Task assigned, Task reassigned, Deadline reminder, Overdue, System.	Should	`notifications.view`	REQ-NOTIF-08
FR-NOTIF-10	The system shall filter the list by Read state: All, Unread, Read, and reload from page 1 whenever either filter changes.	Should	`notifications.view`	REQ-NOTIF-09
FR-NOTIF-11	The system shall clear both filters and reload from page 1 when the user clicks Reset or Clear filters.	Should	`notifications.view`	REQ-NOTIF-10
FR-NOTIF-12	The system shall mark one of the user's own notifications as read on Mark as read, and do nothing further when it is already read.	Must	`notifications.manage`	REQ-NOTIF-11
FR-NOTIF-13	The system shall show Mark all as read only to users holding `notifications.manage` and disable it while nothing is unread.	Should	`notifications.manage`	REQ-NOTIF-12
FR-NOTIF-14	The system shall mark all of the user's unread notifications as read on Mark all as read.	Must	`notifications.manage`	REQ-NOTIF-13
FR-NOTIF-15	The system shall, after marking read, remove the unread highlight, update the bell, and reload the list when a Read-state filter is active.	Should	`notifications.manage`	REQ-NOTIF-14
FR-NOTIF-16	The system shall open the related task's details when the user clicks its task code, applying the task's own access rules.	Must	Task access rules	REQ-NOTIF-15
FR-NOTIF-17	The system shall keep a notification whose task has been deleted and show it without a task link.	Should	`notifications.view`	REQ-NOTIF-16
FR-NOTIF-18	The system shall offer no way to delete a notification; notifications can only be marked as read.	Must	–	REQ-NOTIF-17, BR-NOTIF-02

ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-19	The system shall show "No notifications yet – Tasks assigned or reassigned to you will appear here." when the user has none, "No notifications match these filters – Try a different type or clear the filters." with Clear filters when filters match nothing, and "Could not load notifications. Check your connection and try again." with Retry when loading fails.	Should	`notifications.view`	REQ-NOTIF-18
FR-NOTIF-20	The system shall never show or change another user's notification and shall treat such a notification as not found.	Must	`notifications.manage`	REQ-NOTIF-19, BR-NOTIF-01, NFR-NOTIF-07
FR-NOTIF-21	The system shall show "Could not mark the notification as read." or "Could not mark all notifications as read." when marking fails, including for a user without `notifications.manage`.	Should	`notifications.manage`	REQ-NOTIF-11, REQ-NOTIF-13

Use case: UC-NOTIF-01 – View and filter notifications

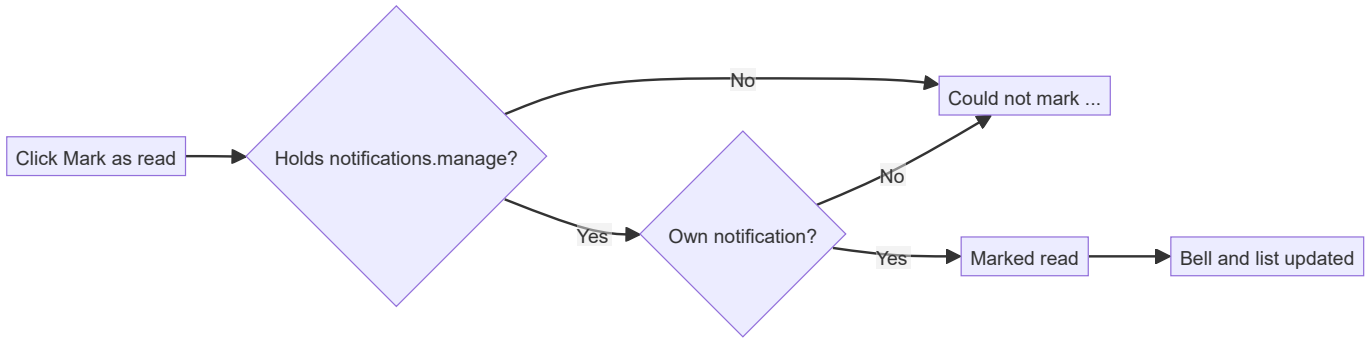
Item	Detail
Actor	Any user with `notifications.view`
Preconditions	Signed in
Trigger	Clicks the bell or Communication Management → Notifications
Main flow	1. System checks `notifications.view` (FR-NOTIF-01). 2. System loads the newest 20 own notifications and the unread state (FR-NOTIF-04, 05). 3. User picks Type and/or Read state; list reloads from page 1 (FR-NOTIF-09, 10). 4. User pages with Previous / Next.
Alternate flows	A1: Reset / Clear filters → all notifications again (FR-NOTIF-11).
Exceptions	E1: no permission → dashboard. E2: none / no match / load error → FR-NOTIF-19 states.
Postconditions	Nothing changes; the user has seen their notifications



Use case: UC-NOTIF-02 – Mark notifications as read

Item	Detail
Actor	User with `notifications.manage`
Preconditions	Notification Center open with unread rows
Trigger	Clicks Mark as read on a row, or Mark all as read
Main flow	1. System checks the key and ownership (FR-NOTIF-12, 14, 20). 2. Rows lose the highlight; the bell dot clears when nothing is unread (FR-NOTIF-15). 3. With a Read-state filter, the list reloads.

Item	Detail
Alternate flows	A1: row already read → no change (FR-NOTIF-12).
Exceptions	E1: no key or foreign notification → "Could not mark the notification as read." / "Could not mark all notifications as read." (FR-NOTIF-21).
Postconditions	Notifications read; none deleted (FR-NOTIF-18)



Use case: UC-NOTIF-03 – Open the related task

Item	Detail
Actor	User with `notifications.view`
Preconditions	A row with a task code
Trigger	Clicks the task code
Main flow	1. System opens the task details screen (FR-NOTIF-16). 2. The task screen applies its own access rules.
Alternate flows	A1: task deleted → no link shown (FR-NOTIF-17).
Exceptions	E1: user may not see that task → handled by the task screen (Task Management FRD).
Postconditions	Task details shown; the notification's read state is unchanged

Field validations

Field	Rule	Error message
Page size	1–100; otherwise 20 is used	None
Type filter	One of the listed types or All types	None (drop-down)
Read state filter	All, Unread or Read	None (drop-down)
Notification id (mark read)	Must belong to the caller	"Could not mark the notification as read."

Business rules applied: BR-NOTIF-01, BR-NOTIF-02; global rules on authorised access and private notifications.

Notifications / side effects

Event	Who is notified	Channel
Notification marked read	– (bell dot clears when nothing unread)	In-app

Messages

Situation	Message
Empty list	"No notifications yet – Tasks assigned or reassigned to you will appear here."

Situation	Message
No filter matches	"No notifications match these filters – Try a different type or clear the filters."
Load failed	"Could not load notifications. Check your connection and try again."
Mark one failed	"Could not mark the notification as read."
Mark all failed	"Could not mark all notifications as read."
No permission to open the screen	Returned to the dashboard; no message

4.1.2 Function: Event Notifications (system generated, no screen)

Field	Value
Route / location	No screen; results appear in the Notification Center and on the bell
Access	System; recipients are the affected users
Entry points	Task created, task reassigned, comment added / edited / deleted, worked hours changed (Task Management); MFA reset by an administrator, password reset with the authenticator (Authentication / My Profile)

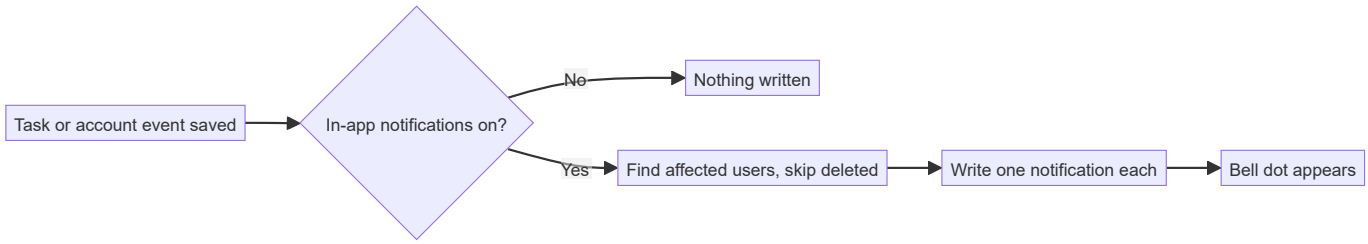
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-22	The system shall notify the assignee when a task is created, with the type "Task assigned" and the text "A task was assigned to you: *task title*".	Must	System	REQ-NOTIF-20
FR-NOTIF-23	The system shall notify the new assignee of a reassignment with the type "Task reassigned" and the text "A task was reassigned to you: *task title*. Reason: *reason*".	Must	System	REQ-NOTIF-21
FR-NOTIF-24	The system shall notify the previous assignee, the recipients and the acting person of a reassignment with "*Name* reassigned task "*title*" (*code*) from *old* to *new*. Reason: *reason*".	Must	System	REQ-NOTIF-21
FR-NOTIF-25	The system shall notify the task's assignee, recipients and the acting person when a comment is added, edited or deleted, with the type Task comment and "*Name added / edited / deleted* a comment on "*title*": *detail*".	Must	System	REQ-NOTIF-22
FR-NOTIF-26	The system shall notify the task's assignee and recipients when the worked hours of a task change, with the type "System", stating what changed and the new total worked.	Should	System	REQ-NOTIF-23
FR-NOTIF-27	The system shall send the account owner a "System" notification "An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile." when an administrator resets their authenticator MFA.	Must	System	REQ-NOTIF-24
FR-NOTIF-28	The system shall send the account owner a "System" notification "Your password was reset with your authenticator app. If this was not you, contact an administrator now." when their password is reset with the authenticator app.	Must	System	REQ-NOTIF-25
FR-NOTIF-29	The system shall write no event notification while in-app notifications are switched off in Application Settings, and shall keep all existing notifications.	Must	System	REQ-NOTIF-26, BR-NOTIF-11
FR-NOTIF-30	The system shall write no notification for a deleted user.	Must	System	REQ-NOTIF-27

ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-31	The system shall keep the task change that caused a notification even when writing the notification fails.	Must	System	BR-NOTIF-13

Use case: UC-NOTIF-04 – Receive an event notification

Item	Detail
Actor	System; affected users
Preconditions	In-app notifications on (Application Settings); affected users not deleted
Trigger	One of the entry-point events above
Main flow	1. Event saved in its own module. 2. System determines the affected users (FR-NOTIF-22 – 28). 3. System writes one notification per user. 4. Each user's bell dot appears within 15 seconds or on the next page change (FR-NOTIF-03).
Alternate flows	A1: toggle off → nothing written; old notifications kept (FR-NOTIF-29).
Exceptions	E1: writing fails → the event itself stays saved (FR-NOTIF-31).
Postconditions	Notifications written for the affected users only



Business rules applied: BR-NOTIF-11, BR-NOTIF-13; global rule that the assigned employee is always included in task notifications.

Notifications / side effects: see §7 (N-01 – N-07). Comment, reassignment and worked-hours events also produce mails (Mail FRD FR-MAIL-25 – FR-MAIL-27).

Messages: the notification texts in FR-NOTIF-22 – FR-NOTIF-28; no user-facing error messages.

4.1.3 Function: Automatic Deadline Reminder Engine (background, no screen)

Field	Value
Route / location	No screen; runs in the background of the application
Access	System; switches in Reminder Settings (`settings.reminders.manage`); per-task fields in Task Management
Entry points	Application start, then every 60 seconds

Functional requirements

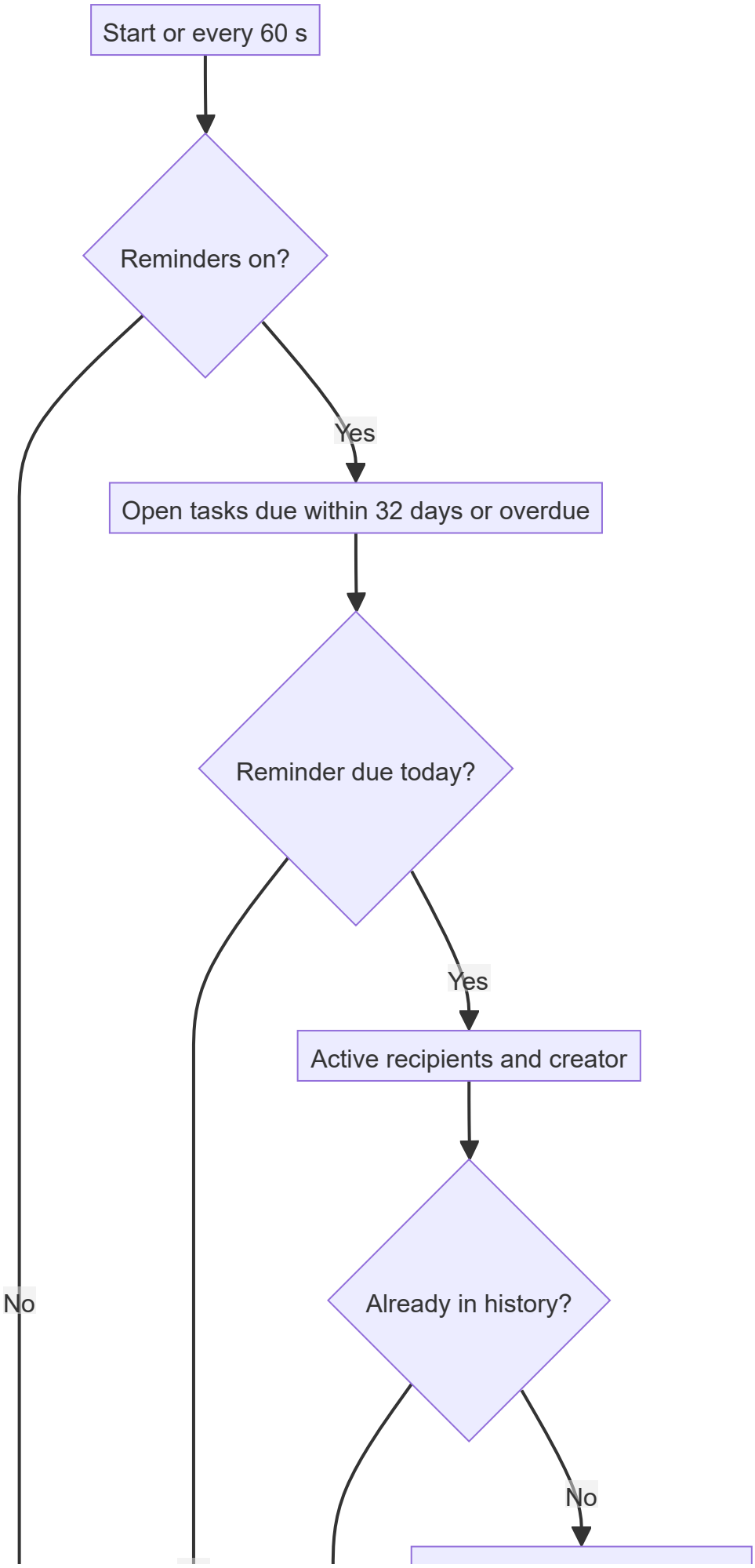
ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-32	The system shall run a reminder pass when the application starts and every 60 seconds after that, without user action.	Must	System	REQ-NOTIF-28, NFR-NOTIF-01
FR-NOTIF-33	The system shall send no reminders while reminders are switched off in Reminder Settings.	Must	System	REQ-NOTIF-29, BR-NOTIF-11
FR-NOTIF-34	The system shall consider only open tasks (not completed, not cancelled, not deleted) whose deadline is at most 32 days ahead or already passed.	Must	System	REQ-NOTIF-30, BR-NOTIF-03
FR-NOTIF-35	The system shall decide the current day and the days until the deadline in the application timezone, using UTC when no timezone is set, and count days correctly across daylight-saving changes.	Must	System	REQ-NOTIF-31, BR-NOTIF-07, NFR-NOTIF-06
FR-NOTIF-36	The system shall send the first reminder on the day that is exactly the task's first-reminder days before the deadline day.	Must	System	REQ-NOTIF-32
FR-NOTIF-37	The system shall send the second reminder on the day that is exactly the task's second-reminder days before the deadline day.	Must	System	REQ-NOTIF-33
FR-NOTIF-38	The system shall send only one reminder (the second reminder) on a day that matches both the first and the second reminder days.	Should	System	REQ-NOTIF-34
FR-NOTIF-39	The system shall send a due-today reminder on the deadline day, only while the deadline time has not passed, when the deadline-day reminder is on in Reminder Settings or the task's first or second reminder days is 0.	Must	System	REQ-NOTIF-35, BR-NOTIF-08
FR-NOTIF-40	The system shall send an overdue reminder, while the task stays open, every day from the day after the deadline day when the overdue frequency is Daily, or every 7th day after the deadline day (7, 14, 21 … days) when it is Weekly.	Must	System	REQ-NOTIF-36, BR-NOTIF-09
FR-NOTIF-41	The system shall send each reminder to the task's notification recipients (always including the assignee) and the task creator, each person once, active users only.	Must	System	REQ-NOTIF-37, BR-NOTIF-04
FR-NOTIF-42	The system shall deliver each reminder (task, person, reminder type, day) at most once, however often the engine runs, overlaps or restarts.	Must	System	REQ-NOTIF-38, BR-NOTIF-05, NFR-NOTIF-04

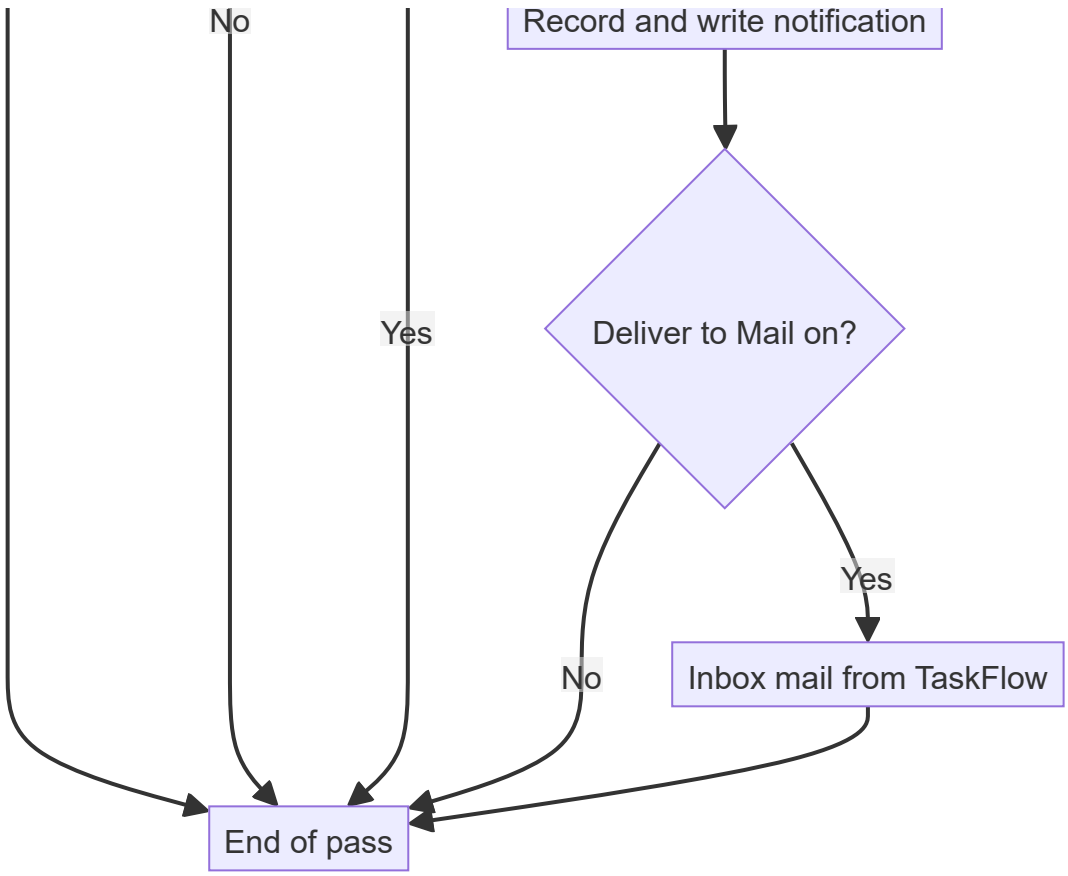
ID	Requirement	Priority	Permission	BRD trace
FR-NOTIF-43	The system shall send only reminders due on the current day and never send a reminder for a day that has ended.	Must	System	REQ-NOTIF-39, BR-NOTIF-06
FR-NOTIF-44	The system shall write nothing for a task that is completed, cancelled, deleted or given a new deadline while the pass is running.	Must	System	REQ-NOTIF-40
FR-NOTIF-45	The system shall use the texts "Reminder: "*title*" is due in *N* days (*dd MMM yyyy, HH:mm*)." ("1 day" when N is 1) for first and second reminders, "Reminder: "*title*" is due today at *HH:mm*." for due-today and "Overdue: "*title*" was due on *dd MMM yyyy, HH:mm*." for overdue.	Must	System	REQ-NOTIF-41
FR-NOTIF-46	The system shall give first, second and due-today reminders the type "Deadline reminder" and overdue reminders the type "Overdue".	Should	System	REQ-NOTIF-42
FR-NOTIF-47	The system shall, while "Also deliver reminders to Mail" is on, also place each reminder in the recipient's Inbox as a mail from "TaskFlow" with the subject "Task reminder: ", "Task due today: " or "Overdue task: " followed by the title, and a body with the reminder text, an "Open the task" link and "This is an automatic reminder from TaskFlow."; the in-app notification is always written.	Should	System	REQ-NOTIF-43, BR-NOTIF-12
FR-NOTIF-48	The system shall record every delivered reminder in the reminder history with task, recipient, reminder type (First, Second, Due today, Overdue), reminder day, status "Sent" and processing time.	Must	System	REQ-NOTIF-44
FR-NOTIF-49	The system shall log a failed pass and run the next pass at the next interval.	Should	System	REQ-NOTIF-45, NFR-NOTIF-05
FR-NOTIF-50	The system shall use the task's current deadline, status and reminder fields on every pass, so that a change takes effect from the next pass.	Must	System	REQ-NOTIF-30, REQ-NOTIF-40, BR-NOTIF-10

Use case: UC-NOTIF-05 – Automatic deadline reminder pass

Item	Detail
Actor	System (reminder engine); recipients (assignee, task recipients, creator)
Preconditions	Application running; reminders on in Reminder Settings
Trigger	Application start or 60-second interval

Item	Detail
Main flow	1. Pass starts (FR-NOTIF-32). 2. Reminders on? (FR-NOTIF-33). 3. Find open candidate tasks (FR-NOTIF-34). 4. Decide today's reminder per task in the application timezone (FR-NOTIF-35 – 40). 5. Resolve active recipients (FR-NOTIF-41). 6. Skip reminders already in the history (FR-NOTIF-42). 7. Record, write the notification and, if on, the Inbox mail (FR-NOTIF-45 – 48).
Alternate flows	A1: first and second days equal → one reminder (FR-NOTIF-38). A2: Mail delivery off → notification only.
Exceptions	E1: task changed during the pass → nothing written (FR-NOTIF-44). E2: pass fails → logged; next pass retries (FR-NOTIF-49). E3: application stopped for a whole day → that day is not caught up (FR-NOTIF-43).
Postconditions	Each due reminder delivered exactly once and recorded


Start or every 60 s
Reminders on?
Yes
No
Open tasks due within 32 days or overdue
Reminder due today?
Yes
Active recipients and creator
Already in history?
No



Worked example (defaults: first 2 days, second 1 day, overdue Daily, deadline-day reminder on) – task "Prepare Monthly Report", deadline 15-Oct-2026 17:00, assignee Rahul, creator Anita, recipient Priya:

Day	Reminder type	Text (each of Rahul, Anita, Priya, once)
13-Oct-2026	First	Reminder: "Prepare Monthly Report" is due in 2 days (15 Oct 2026, 17:00).
14-Oct-2026	Second	Reminder: "Prepare Monthly Report" is due in 1 day (15 Oct 2026, 17:00).
15-Oct-2026, before 17:00	Due today	Reminder: "Prepare Monthly Report" is due today at 17:00.
16-Oct-2026 onward	Overdue	Overdue: "Prepare Monthly Report" was due on 15 Oct 2026, 17:00.
After completion	–	None

Field validations

Field	Rule	Error message
Task reminder days (read from the task)	0–30, set in Task Management	Validated in the Task Management FRD
Overdue frequency (read from the task)	Daily or Weekly	Validated in the Task Management FRD

Business rules applied: BR-NOTIF-03 – BR-NOTIF-10, BR-NOTIF-12; global rules: overdue definition, completed and cancelled tasks get no reminders, configurable reminder settings, duplicate prevention, one timezone, Mail failure does not stop in-app notifications.

Notifications / side effects: see §7 (N-08 – N-11) and the Mail FRD §5.1.

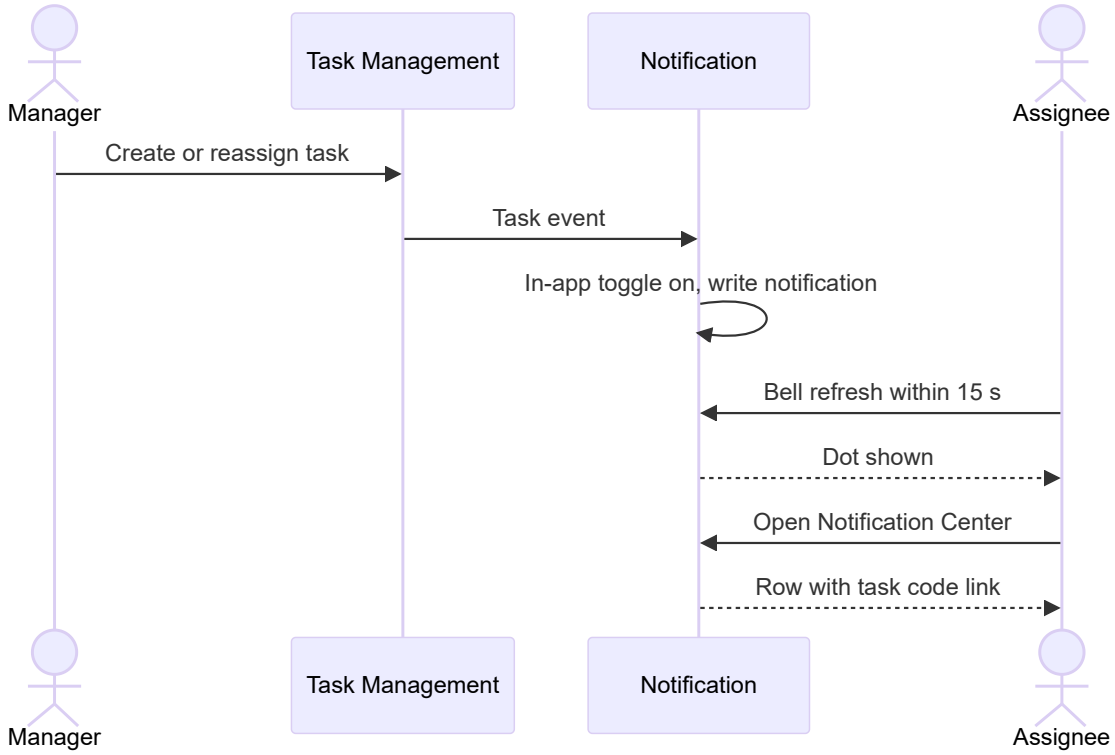
Messages: the reminder texts in FR-NOTIF-45; no user-facing error messages (failures are logged).

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5. Cross-Module Workflows

5.1 Task created or reassigned

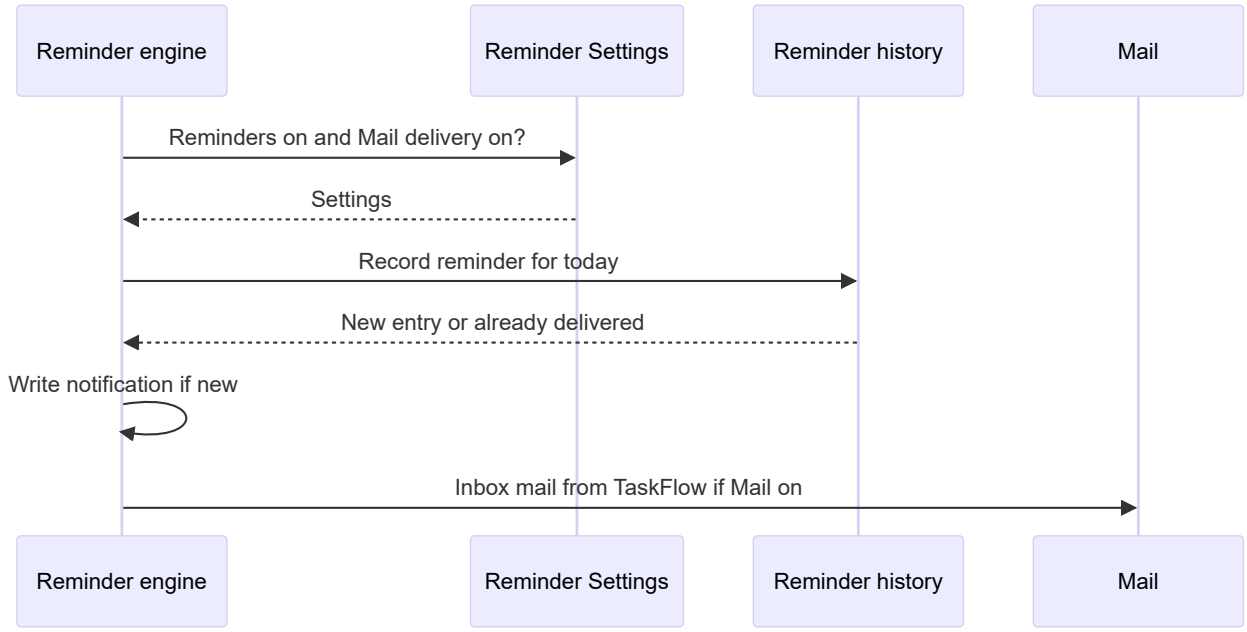
Step	Module	Actor / System	Action	FR reference
1	TASK	Manager	Creates or reassigns a task	Task Management FRD
2	APP	System	Checks the in-app notifications toggle	FR-NOTIF-29
3	NOTIF	System	Writes "Task assigned" / "Task reassigned" notifications	FR-NOTIF-22, FR-NOTIF-23, FR-NOTIF-24
4	MAIL	System	Reassignment also sends a mail from the acting person	FR-MAIL-26 (Mail FRD)
5	NOTIF	Assignee	Sees the bell dot, opens the Notification Center and the task	FR-NOTIF-03, FR-NOTIF-16



5.2 Deadline reminder with Mail delivery

Step	Module	Actor / System	Action	FR reference
1	NOTIF	System	Pass starts; reads Reminder Settings	FR-NOTIF-32, FR-NOTIF-33
2	NOTIF	System	Finds due reminders and active recipients	FR-NOTIF-34 – FR-NOTIF-41
3	NOTIF	System	Records each reminder once and writes the notification	FR-NOTIF-42, FR-NOTIF-48
4	MAIL	System	If "Also deliver reminders to Mail" is on, places the Inbox mail from "TaskFlow"	FR-NOTIF-47, FR-MAIL-21

Step	Module	Actor / System	Action	FR reference
5	NOTIF / MAIL	Recipient	Reads the notification or mail and opens the task	FR-NOTIF-16, FR-MAIL-12



6. Reports & Exports

Not applicable for this sub-module.

7. Notifications & Messages Catalogue

Code	Trigger	Recipients	Channel	Text
N-01	Task created	Assignee	In-app (Task assigned)	"A task was assigned to you: *task title*"
N-02	Task reassigned	New assignee	In-app (Task reassigned)	"A task was reassigned to you: *task title*. Reason: *reason*"
N-03	Task reassigned	Previous assignee, recipients, acting person	In-app (Task reassigned)	"*Name* reassigned task "*title*" (*code*) from *old* to *new*. Reason: *reason*"
N-04	Comment added / edited / deleted	Assignee, recipients, acting person	In-app (Task comment); Mail if toggle on	"*Name added / edited / deleted* a comment on "*title*": *detail*"
N-05	Worked hours changed	Assignee, recipients	In-app (System)	What changed and the new total worked
N-06	Administrator resets authenticator MFA	Account owner	In-app (System)	"An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."

Code	Trigger	Recipients	Channel	Text
N-07	Password reset with the authenticator	Account owner	In-app (System)	"Your password was reset with your authenticator app. If this was not you, contact an administrator now."
N-08	First reminder day	Reminder recipients	In-app (Deadline reminder); Mail "Task reminder: *title*" if on	"Reminder: "*title*" is due in *N* days (*dd MMM yyyy, HH:mm*)."
N-09	Second reminder day	Reminder recipients	In-app (Deadline reminder); Mail "Task reminder: *title*" if on	As N-08 ("1 day" when N is 1)
N-10	Deadline day, before the deadline time	Reminder recipients	In-app (Deadline reminder); Mail "Task due today: *title*" if on	"Reminder: "*title*" is due today at *HH:mm*."
N-11	Each Daily / Weekly overdue day	Reminder recipients	In-app (Overdue); Mail "Overdue task: *title*" if on	"Overdue: "*title*" was due on *dd MMM yyyy, HH:mm*."

Screen messages are listed in §4.1.1.

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8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Notification	One alert for one user: recipient, related task (optional), type (Task assigned, Task reassigned, Deadline reminder, Overdue, System, Task comment), message, read flag, created time	System	Kept permanently; only the read flag changes; removed only together with its user or task
Reminder history	Proof of a delivered reminder: task, recipient, reminder type (First, Second, Due today, Overdue), reminder day, status, processed time	System (reminder engine)	Unique per task, recipient, reminder type and day; status "Sent"
Task reminder fields (read only)	Deadline, status, first and second reminder days, overdue frequency, creator, notification recipients	Task Management	Owned and validated by Task Management

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9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-NOTIF-01	Engine runs every 60 seconds; a day's reminder goes out on the first pass of that day	Reminder engine (FR-NOTIF-32)
NFR-NOTIF-02	Bell refreshed on page change and at least every 15 seconds	Header bell (FR-NOTIF-03)
NFR-NOTIF-03	20 notifications per page, at most 100	Notification Center (FR-NOTIF-05, FR-NOTIF-06)
NFR-NOTIF-04	No duplicates across overlapping passes or restarts	Reminder engine (FR-NOTIF-42)

ID	Requirement	Affects
NFR-NOTIF-05	A failed pass is logged and the next pass runs	Reminder engine (FR-NOTIF-49)
NFR-NOTIF-06	Day boundaries correct across daylight-saving changes	Reminder engine (FR-NOTIF-35)
NFR-NOTIF-07	Foreign notifications reported as not found	Notification Center (FR-NOTIF-20)
NFR-NOTIF-08	Reminder switches and defaults edited in Reminder Settings; run interval set in configuration	Reminder engine (FR-NOTIF-32, FR-NOTIF-33)

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	The application runs continuously; a day's reminders are sent while it runs on that day.	Jojo George
Assumption	One application timezone applies to all users; single-server deployment with one reminder engine.	Jojo George
Open issue	⚠️ To be confirmed: comment notifications have their own type (Task comment), but the Type filter and type chip do not list it, so these rows show an empty chip and cannot be filtered. Confirm whether "Task comment" should be added to FR-NOTIF-09.	Jojo George
Open issue	⚠️ To be confirmed: the per-row Mark as read button is shown to every viewer although marking needs `notifications.manage`. Confirm whether it should be hidden like Mark all as read.	Jojo George
Open issue	⚠️ To be confirmed: deadline reminders are still written when in-app notifications are switched off in Application Settings (only event notifications follow that toggle, FR-NOTIF-29). Confirm the intended behaviour.	Jojo George
Open issue	⚠️ To be confirmed: with the Weekly overdue frequency the first overdue reminder comes 7 days after the deadline day (FR-NOTIF-40), not on the day after. Confirm this is intended.	Jojo George
Open issue	Notifications are never deleted; clean-up is a later phase (BRD R-NOTIF-02).	Jojo George

11. Requirements Traceability Matrix

Test case column: automated test suites from the TaskFlow test projects (backend test classes and frontend `*.spec.ts` files). The mapping was checked against the test sources; per-test mapping belongs in the LLD. "Manual UI check" = no automated test covers the behaviour.

BRD requirements → FRD

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
REQ-NOTIF-01	FR-NOTIF-01	Notification Center	UC-NOTIF-01	`NotificationsEndpointsTests`

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
REQ-NOTIF-02	FR-NOTIF-02	Header bell	UC-NOTIF-01	`shell.component.spec.ts`
REQ-NOTIF-03	FR-NOTIF-03	Header bell	UC-NOTIF-01, UC-NOTIF-04	`NotificationsEndpointsTests`, `shell.component.spec.ts`
REQ-NOTIF-04	FR-NOTIF-04	Notification Center	UC-NOTIF-01	`NotificationServiceTests`
REQ-NOTIF-05	FR-NOTIF-05, FR-NOTIF-06	Notification Center	UC-NOTIF-01	`NotificationServiceTests`
REQ-NOTIF-06	FR-NOTIF-07	Notification Center	UC-NOTIF-01	`NotificationServiceTests`, `notification-center.component.spec.ts`
REQ-NOTIF-07	FR-NOTIF-08	Notification Center	UC-NOTIF-01	Manual UI check
REQ-NOTIF-08	FR-NOTIF-09	Notification Center	UC-NOTIF-01	`NotificationServiceTests`, `notification-center.component.spec.ts`
REQ-NOTIF-09	FR-NOTIF-10	Notification Center	UC-NOTIF-01	`NotificationServiceTests`, `notification-center.component.spec.ts`
REQ-NOTIF-10	FR-NOTIF-11	Notification Center	UC-NOTIF-01	`notification-center.component.spec.ts`
REQ-NOTIF-11	FR-NOTIF-12, FR-NOTIF-21	Notification Center	UC-NOTIF-02	`NotificationServiceTests`, `notification-center.component.spec.ts`
REQ-NOTIF-12	FR-NOTIF-13	Notification Center	UC-NOTIF-02	Manual UI check
REQ-NOTIF-13	FR-NOTIF-14, FR-NOTIF-21	Notification Center	UC-NOTIF-02	`NotificationServiceTests`, `notification-center.component.spec.ts`
REQ-NOTIF-14	FR-NOTIF-15	Notification Center	UC-NOTIF-02	`notification-center.component.spec.ts`, `notification.service.spec.ts`
REQ-NOTIF-15	FR-NOTIF-16	Notification Center	UC-NOTIF-03	`notification-center.component.spec.ts`
REQ-NOTIF-16	FR-NOTIF-17	Notification Center	UC-NOTIF-03	`NotificationServiceTests`
REQ-NOTIF-17	FR-NOTIF-18	Notification Center	UC-NOTIF-02	`NotificationsEndpointsTests`
REQ-NOTIF-18	FR-NOTIF-19	Notification Center	UC-NOTIF-01	`notification-center.component.spec.ts`
REQ-NOTIF-19	FR-NOTIF-20	Notification Center	UC-NOTIF-02	`NotificationServiceTests`
REQ-NOTIF-20	FR-NOTIF-22	Event notifications	UC-NOTIF-04	`TaskManagementServiceTests`
REQ-NOTIF-21	FR-NOTIF-23, FR-NOTIF-24	Event notifications	UC-NOTIF-04	`TaskServiceTests`
REQ-NOTIF-22	FR-NOTIF-25	Event notifications	UC-NOTIF-04	`TaskCommentActivityNotificationTests`

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
REQ-NOTIF-23	FR-NOTIF-26	Event notifications	UC-NOTIF-04	`TaskWorkedHoursNotificationTests`
REQ-NOTIF-24	FR-NOTIF-27	Event notifications	UC-NOTIF-04	`MfaServiceTests`, `MfaAdminResetEndpointsTests`
REQ-NOTIF-25	FR-NOTIF-28	Event notifications	UC-NOTIF-04	`PasswordRecoveryServiceTests`
REQ-NOTIF-26	FR-NOTIF-29	Event notifications	UC-NOTIF-04	`NotificationPublisherTests`, `TaskCommentActivityNotificationTests`
REQ-NOTIF-27	FR-NOTIF-30	Event notifications	UC-NOTIF-04	`ReminderServiceTests`
REQ-NOTIF-28	FR-NOTIF-32	Reminder engine	UC-NOTIF-05	`ReminderSchedulerServiceTests`
REQ-NOTIF-29	FR-NOTIF-33	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-30	FR-NOTIF-34, FR-NOTIF-50	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-31	FR-NOTIF-35	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-32	FR-NOTIF-36	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-33	FR-NOTIF-37	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-34	FR-NOTIF-38	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-35	FR-NOTIF-39	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-36	FR-NOTIF-40	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-37	FR-NOTIF-41	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-38	FR-NOTIF-42	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`
REQ-NOTIF-39	FR-NOTIF-43	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
REQ-NOTIF-40	FR-NOTIF-44, FR-NOTIF-50	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`
REQ-NOTIF-41	FR-NOTIF-45	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-42	FR-NOTIF-46	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-43	FR-NOTIF-47	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
REQ-NOTIF-44	FR-NOTIF-48	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
REQ-NOTIF-45	FR-NOTIF-49	Reminder engine	UC-NOTIF-05	`ReminderSchedulerServiceTests`

BRD business rules and NFRs → FRD

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
BR-NOTIF-01	FR-NOTIF-04, FR-NOTIF-20	Notification Center	UC-NOTIF-01, UC-NOTIF-02	`NotificationServiceTests`
BR-NOTIF-02	FR-NOTIF-18	Notification Center	UC-NOTIF-02	`NotificationsEndpointsTests`
BR-NOTIF-03	FR-NOTIF-34	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
BR-NOTIF-04	FR-NOTIF-41	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
BR-NOTIF-05	FR-NOTIF-42	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`
BR-NOTIF-06	FR-NOTIF-43	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
BR-NOTIF-07	FR-NOTIF-35	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
BR-NOTIF-08	FR-NOTIF-39	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
BR-NOTIF-09	FR-NOTIF-40	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
BR-NOTIF-10	FR-NOTIF-50	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`
BR-NOTIF-11	FR-NOTIF-29, FR-NOTIF-33	Event notifications, Reminder engine	UC-NOTIF-04, UC-NOTIF-05	`NotificationPublisherTests`, `ReminderServiceTests`
BR-NOTIF-12	FR-NOTIF-47	Reminder engine	UC-NOTIF-05	`ReminderServiceTests`
BR-NOTIF-13	FR-NOTIF-31	Event notifications	UC-NOTIF-04	`TaskWorkedHoursNotificationTests` (mail failure only); no test fails a notification during a task change (see LLD §11)

BRD item	FRD requirement(s)	Screen / function	Use case	Test case
NFR-NOTIF-01	FR-NOTIF-32	Reminder engine	UC-NOTIF-05	`ReminderSchedulerServiceTests`
NFR-NOTIF-02	FR-NOTIF-03	Header bell	UC-NOTIF-01	`shell.component.spec.ts`
NFR-NOTIF-03	FR-NOTIF-05, FR-NOTIF-06	Notification Center	UC-NOTIF-01	`NotificationServiceTests`
NFR-NOTIF-04	FR-NOTIF-42	Reminder engine	UC-NOTIF-05	`ReminderIdempotencyTests`
NFR-NOTIF-05	FR-NOTIF-49	Reminder engine	UC-NOTIF-05	`ReminderSchedulerServiceTests`
NFR-NOTIF-06	FR-NOTIF-35	Reminder engine	UC-NOTIF-05	`ReminderCalculatorTests`
NFR-NOTIF-07	FR-NOTIF-20	Notification Center	UC-NOTIF-02	`NotificationServiceTests`
NFR-NOTIF-08	FR-NOTIF-32, FR-NOTIF-33	Reminder engine	UC-NOTIF-05	`SettingsEndpointsTests` (Reminder Settings)

BRD acceptance criteria → FRD

Acceptance criterion	FRD requirement(s)
AC-NOTIF-01	FR-NOTIF-01, FR-NOTIF-02
AC-NOTIF-02	FR-NOTIF-03, FR-NOTIF-15
AC-NOTIF-03	FR-NOTIF-04 – FR-NOTIF-08
AC-NOTIF-04	FR-NOTIF-09, FR-NOTIF-10, FR-NOTIF-11
AC-NOTIF-05	FR-NOTIF-12, FR-NOTIF-13, FR-NOTIF-14
AC-NOTIF-06	FR-NOTIF-16, FR-NOTIF-17, FR-NOTIF-18
AC-NOTIF-07	FR-NOTIF-19, FR-NOTIF-20
AC-NOTIF-08	FR-NOTIF-22 – FR-NOTIF-26
AC-NOTIF-09	FR-NOTIF-27, FR-NOTIF-28
AC-NOTIF-10	FR-NOTIF-29, FR-NOTIF-30
AC-NOTIF-11	FR-NOTIF-32 – FR-NOTIF-40
AC-NOTIF-12	FR-NOTIF-41
AC-NOTIF-13	FR-NOTIF-42, FR-NOTIF-43
AC-NOTIF-14	FR-NOTIF-34, FR-NOTIF-44, FR-NOTIF-50
AC-NOTIF-15	FR-NOTIF-45, FR-NOTIF-46, FR-NOTIF-47

Acceptance criterion	FRD requirement(s)
AC-NOTIF-16	FR-NOTIF-48, FR-NOTIF-49

Untraced BRD items: 0.

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	To be assigned	Pending	–

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